Beyond the Average Treatment Effect

Static Interventions

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WEBR STATUS
Ready!

Synthetic COVID RCT Data

For our first example, we'll use a synthetic dataset of $n = 1000$ patients from a hypothetical clinical trial for a treatment to decrease intubation and death among patients hospitalized with COVID-19.

- The data is based on a database of over 1,500 patients hospitalized at Weill Cornell Medicine New York Presbyterian Hospital prior to 15 May 2020 with COVID-19 confirmed through PCR.
- To replicate a two-arm randomized clinical trial (RCT), we've simulated a hypothetical treatment variable (`A`) that is randomly assigned for each observation with probability 0.5.
- The outcome of interest (`event`) is intubation or death by 15-days post-hospitalization.
- Treatment is associated with increased survival.
- Baseline covariates include: age, sex, BMI, smoking status, whether the patient required supplemental oxygen within 3 hours of presenting to the emergency department, number of comorbidities, number of relevant symptoms, presence of bilateral infiltrates on chest X-ray, dyspnea, and hypertension.

days	event	age	sex	bmi	smoke	o2	num_comorbid	num_symptoms	bilat	dyspnea	hyper	A	id
15	0	69	Male	32.6	No	No	0	7	Yes	0	0	1	1
7	1	59	Male	30.7	Former Smoker	Yes	5	1	Yes	1	1	0	2
15	0	61	Female	21.5	No	No	4	3	No	1	1	0	3
12	1	49	Male	26.9	No	Yes	1	5	Yes	1	1	0	4
15	0	41	Male	32	No	No	0	5	Yes	1	0	0	5
11	0	96	Female	30.9	Former Smoker	No	3	4	No	1	1	1	6
15	0	81	Female	28.8	No	No	1	5	No	1	1	1	7
15	0	55	Female	23.7	No	No	0	6	Yes	0	0	1	8
1	1	83	Male	21	Former Smoker	Yes	4	3	No	1	1	1	9
7	1	62	Male	25.1	No	Yes	0	2	Yes	1	0	0	10

Population mean outcomes

Our goal is to estimate the average treatment effect (ATE) of the hypothetical treatment on intubation or death:

$$\begin{aligned}\theta &= \mathsf{E}(Y^{A=1} - Y^{A=0}) \\ &= \mathsf{E}(Y^{A=1}) - \mathsf{E}(Y^{A=0})\end{aligned}$$

Notice that the ATE is composed of two parameters:

1. the proportion of patients who receive intubation or die in a hypothetical world where all patients receive treatment ($\mathsf{E}(Y^{A=1})$), and
2. the proportion of patients who receive intubation or die in a hypothetical world where no patients receive treatment ($\mathsf{E}(Y^{A=0})$).

We refer to these parameters as *population mean outcomes*. Let's rewrite the target parameter as a function of interventions using the framework we previously discussed. Let $\mathsf{d}_1(a, h) = 1$ and $\mathsf{d}_0(a, h) = 0$. Then,

$$\theta = \mathsf{E}(Y^{\mathsf{d}_1}) - \mathsf{E}(Y^{\mathsf{d}_0}).$$

Remember, we refer to these interventions as static because they return the same value regardless of their input.

Writing shift functions

So, how do we translate the functions d_1 and d_0 into R code to be used with `lmtp`?

> Policies can be specified using one of two arguments in `lmtp` estimators: `shift` or `shifted`.

If using `shift`, we supply a two-argument function of the form

```
d <- function(data, trt) {
  # Insert code here
}
```

- The first argument should correspond to a dataset containing all the variables in O.
- The second argument should expect a string corresponding to the variable(s) A_t in O.
- This function should return a size n vector with the same class as A_t modified according to d_t.

For d_1 this function would be:

Run Code

```
d1 <- function(data, trt) {
  rep(1, nrow(data))
}
```

If we apply this function to our data we are returned a vector of 1s with the same length as the number of rows in the observed data.

Run Code

```
d1(covid, "A") |> head()
d1(covid, "A") |> length()
```

Question: Which estimator should we use to estimate these parameters?

▶ ✅ Answer

Let's now estimate $\mathsf{E}(Y^{\mathsf{d}_1})$ using `lmtp`.

Run Code

```
A <- "A"
Y <- "event"
W <- c("age", "sex", "bmi", "smoke", "o2",
       "num_comorbid", "num_symptoms", "bilat", "dyspnea", "hyper")

set.seed(34465)

treat <- lmtp_tmle(
  data = covid,
  trt = "A",
  outcome = "event",
  baseline = W,
  outcome_type = "binomial",
  shift = d1,
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = "SL.glm"
)

print(treat)
```

Instead of specifying an intervention using `shift`, we could instead supply a modified version of `data` to the `shifted` argument where the variables A_t are replaced with A_t^d.

Run Code

```
tmp <- covid
tmp$A <- d1(covid, "A")

head(tmp)

set.seed(34465)

treat <- lmtp_tmle(
  data = covid,
  trt = "A",
  outcome = "event",
  baseline = W,
  outcome_type = "binomial",
  shifted = tmp,
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = "SL.glm"
)

print(treat)
```

Question: How can we interpret this result?

▶ **Answer**

`lmtp` objects

A call to an `lmtp` estimator returns a list of class `lmtp`.

Values returned in an `lmtp` object.

Value	Description
estimator	The estimator used.
theta	The estimated population LMTP estimand.
standard_error	The estimated standard error of the LMTP estimand.
low	Lower bound of the 95% confidence interval of the LMTP estimand.
high	Upper bound of the 95% confidence interval of the LMTP estimand.
eif	The estimated, un-centered, influence function of the estimate.
shift	The shift function specifying the treatment policy of interest.

Value	Description
outcome_reg	An $n \times \tau + 1$ matrix of outcome regression predictions. The mean of the first column is used for calculating theta.
density_ratios	An $n \times \tau$ matrix of the estimated, non-cumulative, density ratios.
fits_m	A list the same length as `folds`, containing the fits at each time-point for each fold for the outcome regression.
fits_r	A list the same length as `folds`, containing the fits at each time-point for each fold of density ratio estimation.
outcome_type	The outcome variable type.

Let's inspect some of these values.

Run Code

```
hist(treat$density_ratios[, 1],
     main = "Histogram of density ratios",
     xlab = "Density ratio")
```

Problem 1

Write a function to estimate the population mean outcome if no patients received treatment. Assign the result to an object named `dont_treat`.

Run Code

```
d0 <- # Write your shift function here!
dont_treat <- # Type your code here!
```

▶ ✅ Solution

> `lmtp` already implements `d1` and `d0` as `static_binary_on()` and `static_binary_off()`!

Average treatment effect

With estimates of $\mathsf{E}(Y^{\mathsf{d}_1})$ and $\mathsf{E}(Y^{\mathsf{d}_0})$ we can now calculate the ATE. To do so, we'll use the function `lmtp_contrast()`.

Run Code

```
lmtp_contrast(treat, ref = dont_treat)
```

In addition to the ATE, we can also calculate the causal risk ratio and the causal odds ratio.

Run Code

```
lmtp_contrast(treat, ref = dont_treat, type = "rr")
lmtp_contrast(treat, ref = dont_treat, type = "or")
```

Confirming what we already know, we've estimated the treatment as being associated with decreased intubation or death.

References

Goyal, Parag, Justin J Choi, Laura C Pinheiro, Edward J Schenck, Ruijun Chen, Assem Jabri, Michael J Satlin, et al. 2020. "Clinical Characteristics of Covid-19 in New York City." *New England Journal of Medicine* 382 (24): 2372–74.

Williams, Nicholas, Michael Rosenblum, and Iván Díaz. 2022. "Optimising Precision and Power by Machine Learning in Randomised Trials with Ordinal and Time-to-Event Outcomes with an Application to COVID-19." *Journal of the Royal Statistical Society Series A: Statistics in Society* 185 (4): 2156–78.